Understanding High Blood Pressure (Hypertension)

Patient education sample document. For educational purposes only; not medical advice.

What is hypertension?

Hypertension is a condition in which the force of blood against the artery walls is consistently too high. Blood pressure is written as two numbers: systolic pressure (when the heart beats) over diastolic pressure (when the heart rests between beats). It is measured in millimeters of mercury (mmHg). Hypertension is often called the silent killer because it usually causes no symptoms.

Blood pressure categories (adults)

- Normal: below 120/80 mmHg.
- Elevated: systolic 120 to 129 and diastolic below 80 mmHg.
- Stage 1 hypertension: systolic 130 to 139 or diastolic 80 to 89 mmHg.
- Stage 2 hypertension: systolic 140 or higher, or diastolic 90 or higher.
- Hypertensive crisis: higher than 180/120 mmHg. This requires urgent medical attention, especially with chest pain, shortness of breath, severe headache or vision changes.

Risk factors

- Older age and a family history of hypertension.
- Being overweight or obese.
- A diet high in salt (sodium) and low in potassium.
- Physical inactivity.
- Excess alcohol intake and smoking.
- Chronic stress, diabetes and kidney disease.

Complications

Uncontrolled hypertension damages the arteries and organs. It increases the risk of stroke, heart attack, heart failure, kidney disease, vision loss and cognitive decline.

Lifestyle changes that lower blood pressure

- Follow the DASH diet, which is rich in fruits, vegetables, whole grains and low-fat dairy.
- Limit sodium to less than 2,300 mg per day. An ideal target for most adults is 1,500 mg per day.
- Lose excess weight. Even a loss of 5 kg can lower blood pressure.
- Exercise regularly, for example 30 minutes of brisk walking on most days.
- Limit alcohol and stop smoking.
- Manage stress and get enough sleep.

Medicines

When lifestyle changes are not enough, doctors prescribe medicines. Common classes include ACE inhibitors, angiotensin receptor blockers (ARBs), calcium channel blockers and thiazide diuretics. Many people need more than one medicine. Do not stop blood pressure medicine without talking to your doctor.

Measuring blood pressure at home

Sit quietly for five minutes with your back supported and feet flat on the floor. Rest your arm at heart level and use a correctly sized cuff. Avoid caffeine, exercise and smoking for 30 minutes before measuring. Take two readings one minute apart and record the results.